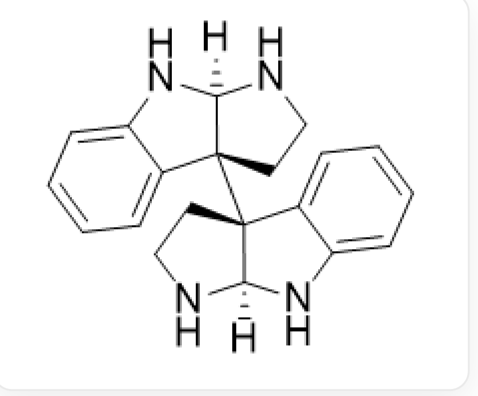
Question

A certain synthetic intermediate $\mathbf{A}$ can be reversibly converted into its more stable isomer $\mathbf{B}$ under KOH-EtOH sealed tube heating conditions. The structure of $\mathbf{A}$ is shown below:



[H][C@@]12NC3=CC=CC=C3[C@]1([C@@]45[C@@](NC6=CC=CC=C64)(NCC5)[H])CCN2

$\mathbf{B}$ reacts with excess iodomethane in the presence of K_2CO_3 as a base, and monomethylation occurs on the two most nucleophilic nitrogen atoms, respectively, to generate product $\mathbf{X}$, whose chemical formula is $C_{22}H_{26}N_4$. It is known that the number of rings of $\mathbf{B}$ and $\mathbf{X}$ are the same as that of $\mathbf{A}$, and no redox reaction occurs during the entire transformation process. $\mathbf{X}$ has two chiral carbon atoms with the same absolute configuration and directly connected to nitrogen atoms. Indicate the number of six-membered rings in $\mathbf{X}$, and the absolute configuration (R or S) of the chiral carbon atoms directly connected to the nitrogen atoms.

A. All other options are not completely correct.

B. 0, S

C. 1, S

D. 2, S

E. 3, S

F. 4, S

G. 5, S

H. 6, S

I. 7, S

J. 8, S

K. 0, R

L. 1, R

M. 2, R

N. 3, R

O. 4, R

P. 5, R

Q. 6, R

R. 7, R

S. 8, R

Answer

Correct Answer: **H**

Detailed Explanation

Compound **A** can be regarded as the product of intramolecular condensation of a highly substituted butanedial with an amine on the substituent, and its properties should be similar to acetals (or hemiacetals).

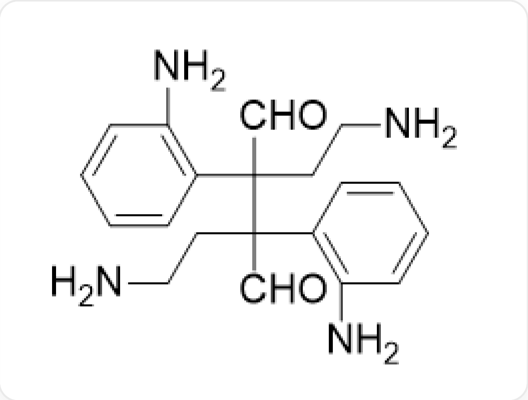
CHECKPOINT 2 PTS

Intramolecular condensation product of butanedial

CHECKPOINT 2 PTS

Properties should be similar to acetals (or hemiacetals)

If we draw the hypothetical un-condensed form of **A**, its structure is as follows (Fischer projection):



NC(C=CC=C1)=C1[C@@](CCN)([C@](CCN)(C2=C(N)C=CC=C2)C=O)C=O

According to the conditions of the question, since the transformation from $\mathbf{A}$ to $\mathbf{B}$ only involves an isomerization reaction, the whole process does not involve redox, and the reaction is reversible, it is likely that a retro-condensation (hydrolysis)-re-condensation reaction occurred, exchanging the amino groups of the acetal, so that all five-membered rings are converted into six-membered rings. This process is energetically favorable, so the equilibrium shifts toward the product $\mathbf{B}$.

CHECKPOINT 3 PTS

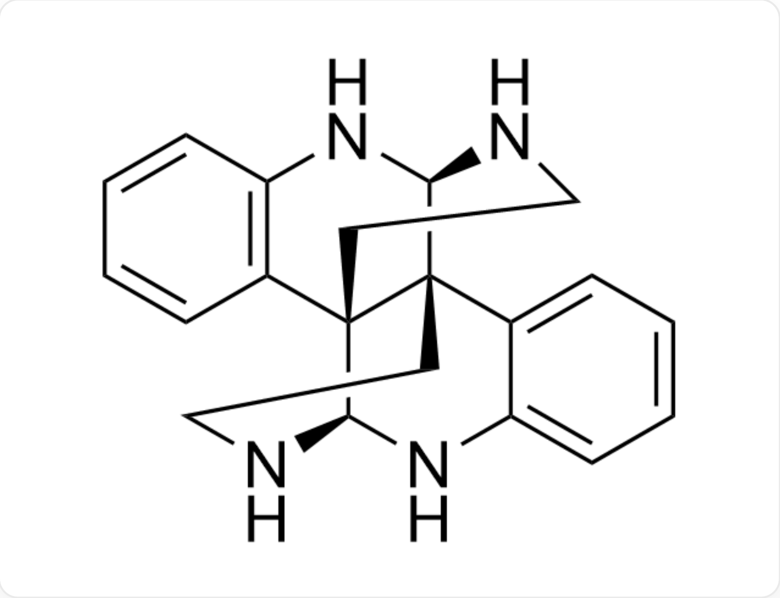
A retro-condensation (hydrolysis)-re-condensation reaction occurred, exchanging the amino groups of the acetal

Due to conformational constraints, each alkylamine can only generate one type of bridged ring structure upon condensation.

CHECKPOINT 3 PTS

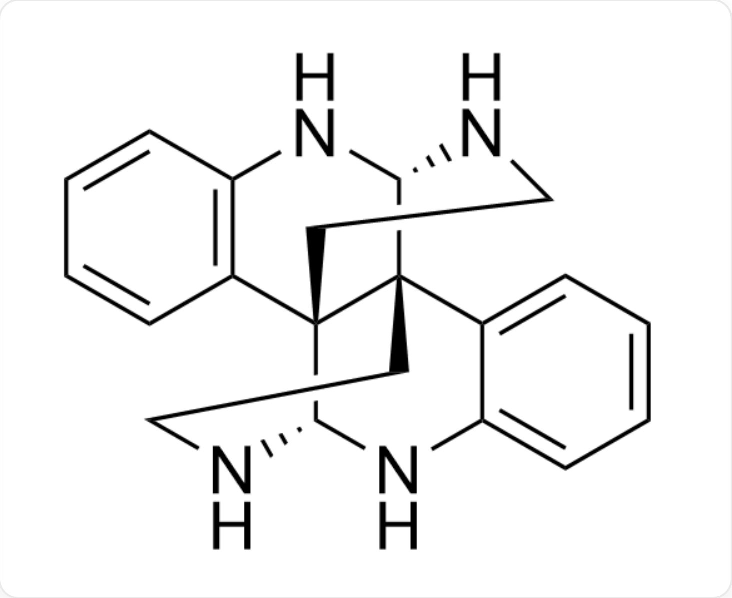
Due to conformational constraints, each alkylamine can only generate one type of bridged ring structure upon condensation

Therefore, the structure of **B** can be deduced as follows:



C12=CC=CC=C1[C@]34[C@@]5(C(C=CC=C6)=C6N[C@H]3NCC5)[C@H](NCC4)N2

Rather than the following structure with too much strain:



C12=CC=CC=C1[C@]34[C@@]5(C(C=CC=C6)=C6N[C@@H]3NCC5)[C@@H](NCC4)N2

CHECKPOINT 3 PTS

The structure of **B** is C12=CC=CC=C1[C@]34[C@@]5(C(C=CC=C6)=C6N[C@H]3NCC5)[C@H](NCC4)N2

For the following methylation, since the basicity of potassium carbonate is low, the amino groups of **B** are all in the unprotonated state in the system.

CHECKPOINT 2 PTS

The basicity of potassium carbonate is low, so the amino groups of **B** are all in the unprotonated state in the system

Obviously, at this time, the nucleophilicity of alkylamines is stronger than that of arylamines.

CHECKPOINT 1 PTS

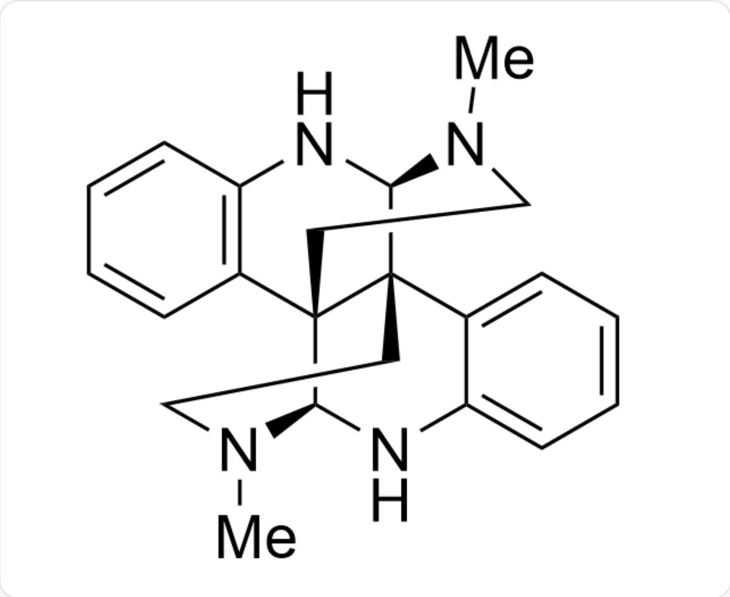
At this time, the nucleophilicity of alkylamines is stronger than that of arylamines

Therefore, methylation occurs on the nitrogens of the two alkylamines.

CHECKPOINT 1 PTS

Methylation occurs on the nitrogens of the two alkylamines

So the structure of **X** is as follows:



CN(CC1)[C@@H]2NC3=C(C=CC=C3)[C@@]41[C@H]5NC6=CC=CC=C6[C@]42CCN5C

CHECKPOINT 2 PTS

Deduce the structure of **X**

Since the alkylamine is methylated, this nitrogen atom is connected to two carbon-containing functional groups, while the arylamine has only one. Therefore, the alkylamine functional group in **X** has a higher priority than the arylamine in the chiral determination, so the symbol of this chiral center is S.

CHECKPOINT 2 PTS

The alkylamine functional group in **X** has a higher priority than the arylamine

At the same time, since $\mathbf{X}$ has six six-membered rings, H is finally selected.

CHECKPOINT 2 PTS

$\mathbf{X}$ has six six-membered rings